WASP-52b

R-1863

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_160906` · created 2026-08-03T09:42:39+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457637.9553 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.062
Transit depth (Rp/Rs)^2	4.63 +/- 2.68 [%]
Orbital Inclination (inc)	88.67 +/- 2.76
Airmass coefficient 1 (a1)	0.3 +/- 0.18
Airmass coefficient 2 (a2)	0.6 +/- 0.36
Scatter in the residuals of the lightcurve fit is	45.63 %
Best Comparison Star	#1 - [451, 407]
Optimal Aperture	1.7
Optimal Annulus	6.25
Transit Duration (day)	0.087 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.062 vs 0.1646 (deviation 0.81σ)
Duration fit vs published	0.087 d vs 0.0758 d (deviation 1.12σ)
Mid-time O–C	6.1 min
Depth SNR	3.5
Residual scatter	45.63 %

Light curve

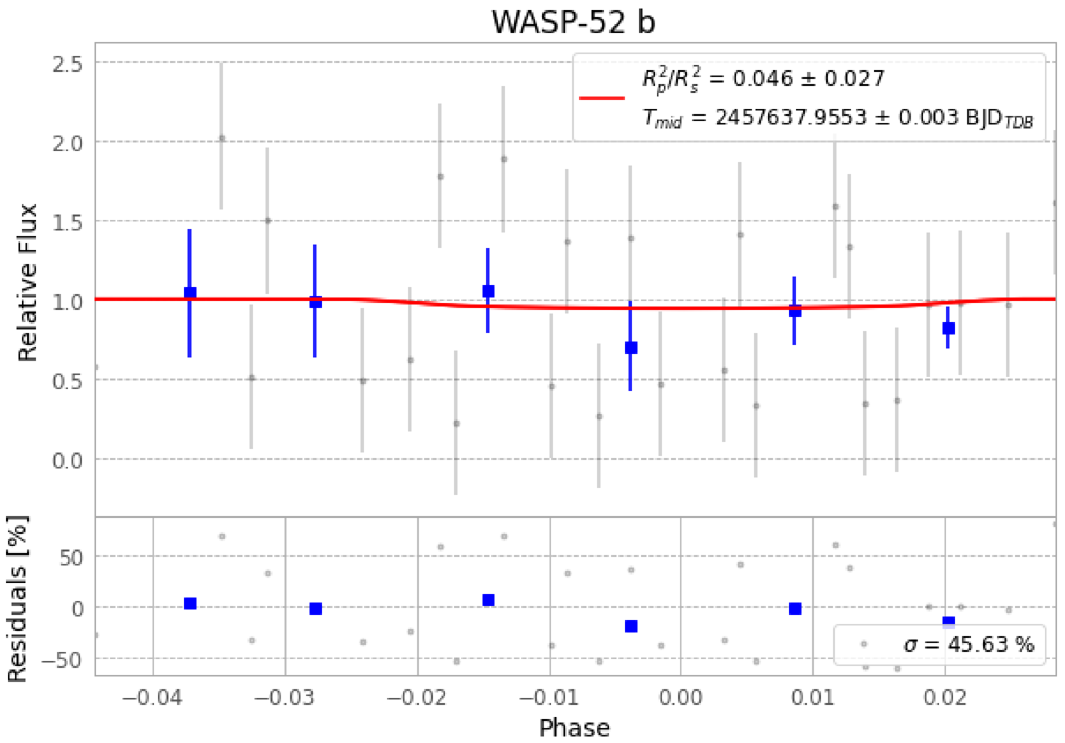

Figure 1 — FinalLightCurve_WASP-52 b_2016-09-06.png (SHA-256 d176427d06a5c8e0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [236, 337], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 017ba3bea6780de4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

62 FITS frames (41 MB), `WASP-52160906085412.FITS` → `WASP-52160906115712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-160906.log (35faf4ce4596…), FinalLightCurve_WASP-52 b_2016-09-06.png (d176427d06a5…), FinalLightCurve_WASP-52 b_2016-09-06.csv (addd791c599d…)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 09:42Z.